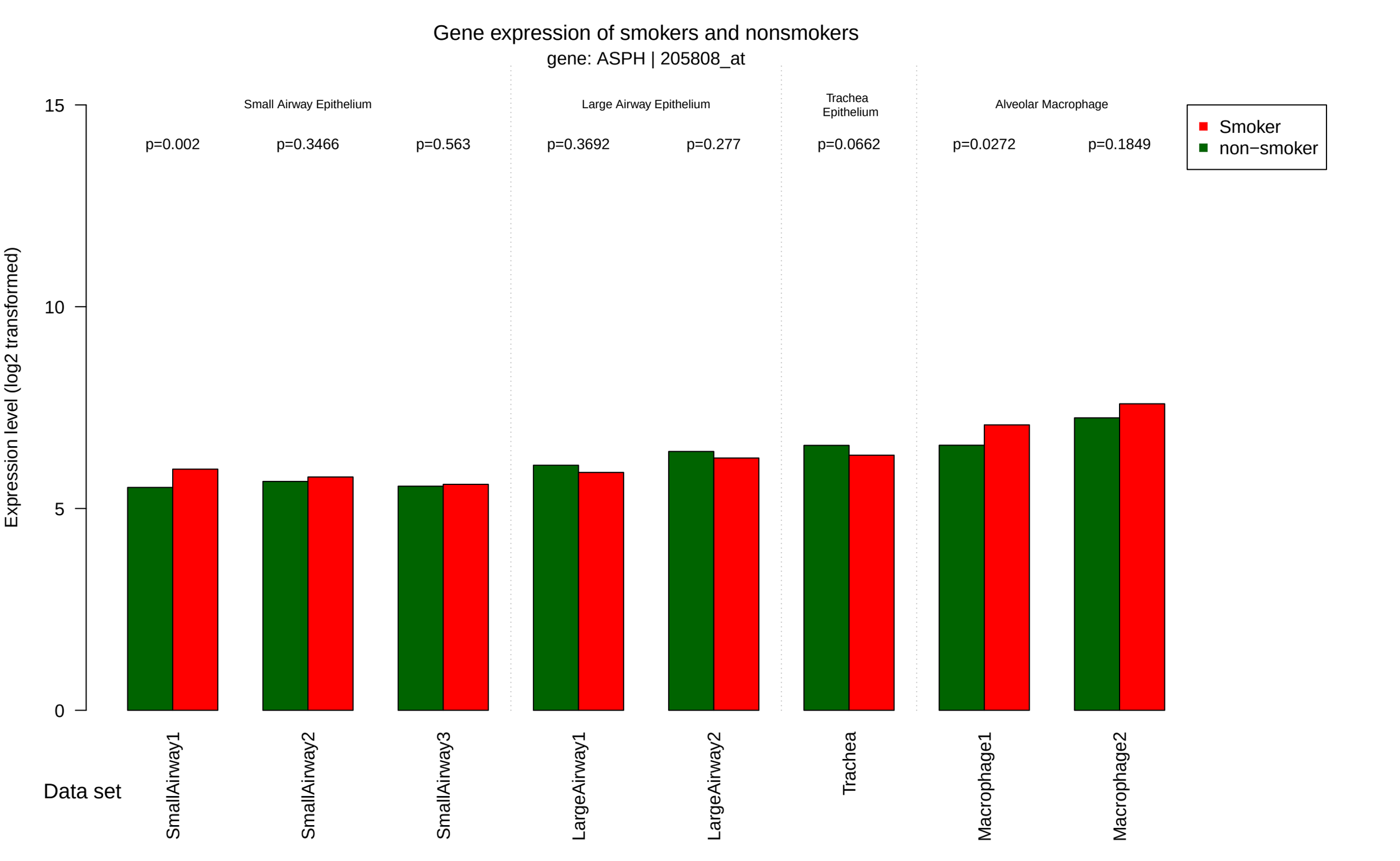
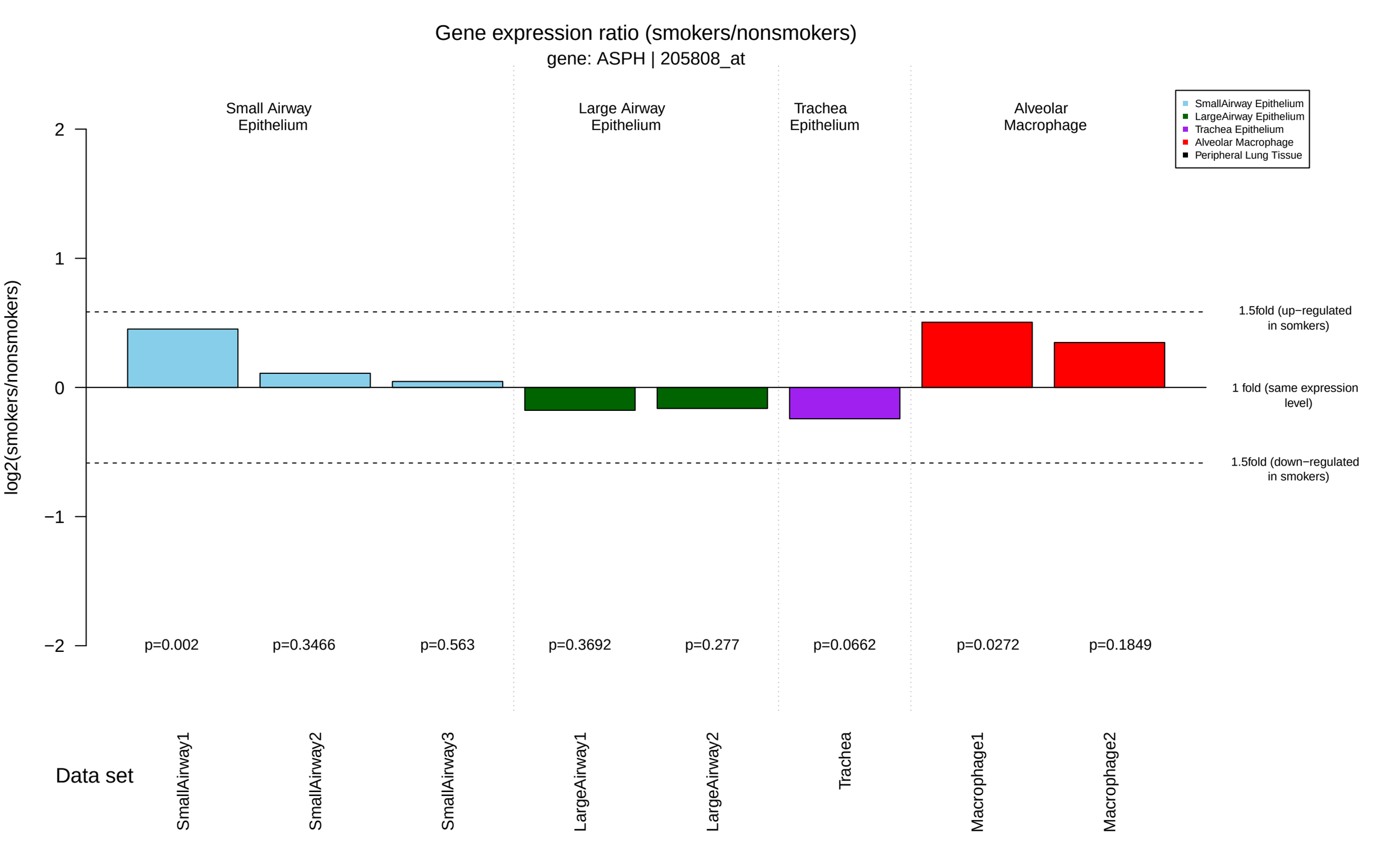
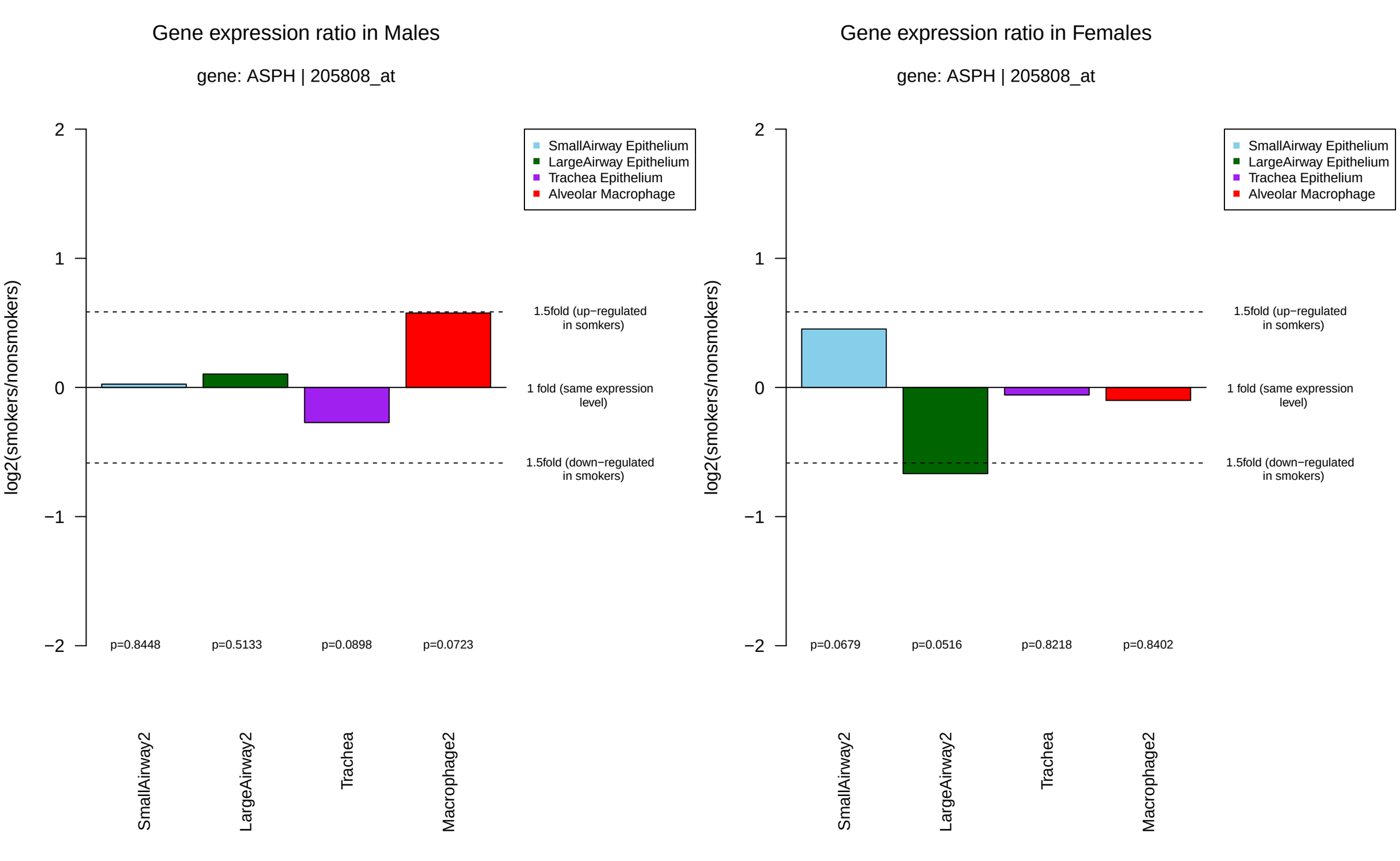
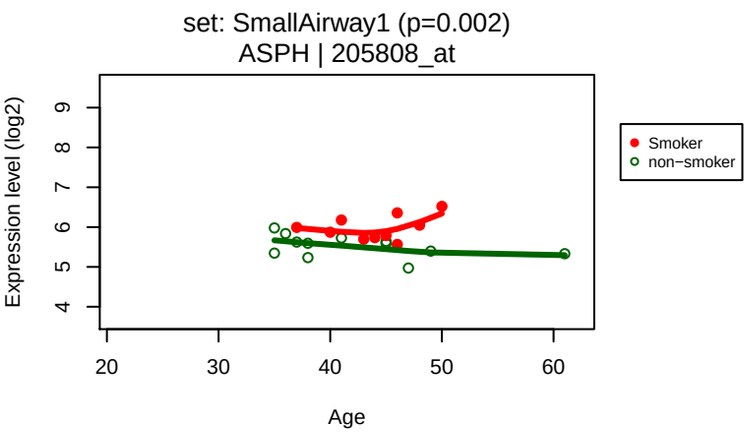
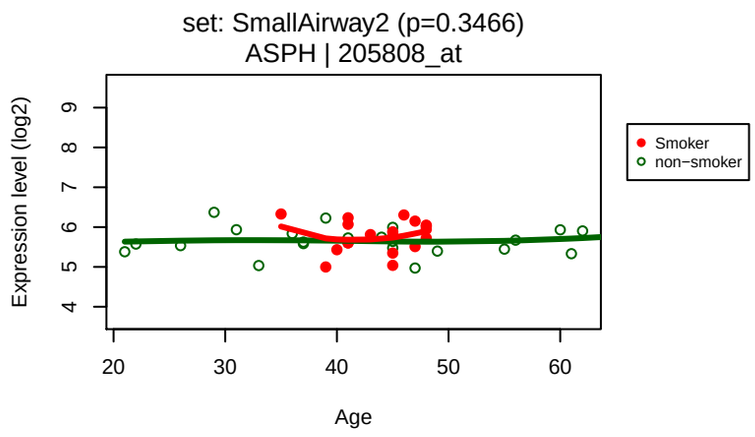
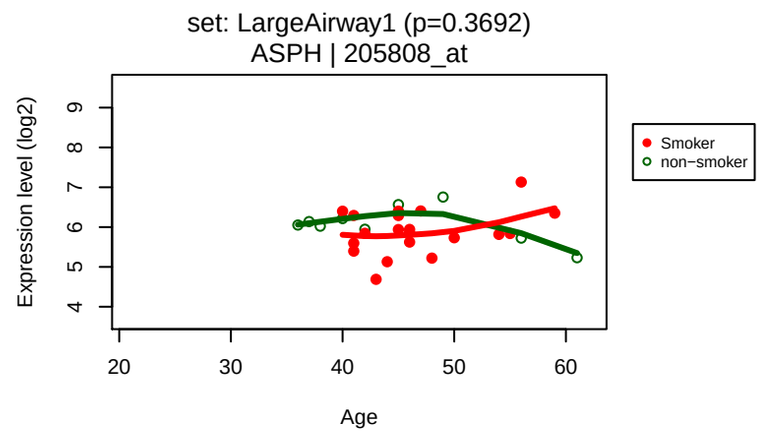

Gene expression of smokers and nonsmokers
gene: ASPH | 205808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.002
p=0.3466
p=0.563
p=0.3692
p=0.277
p=0.0662
p=0.0272
p=0.1849
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

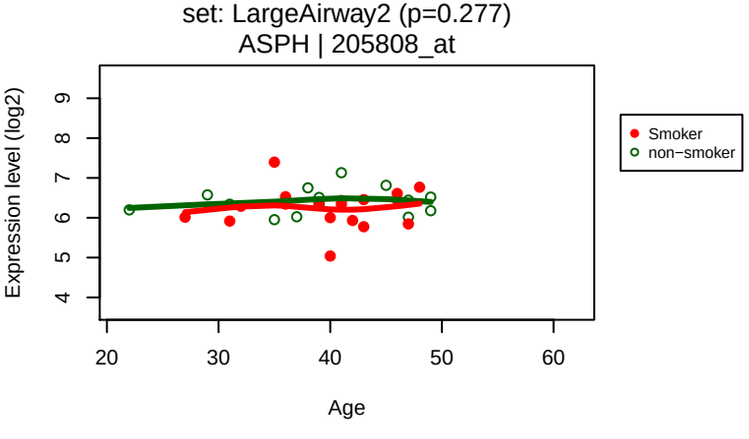

Gene expression ratio (smokers/nonsmokers)
gene: ASPH | 205808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0.3466
p=0.563
p=0.3692
p=0.277
p=0.0662
p=0.0272
p=0.1849
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

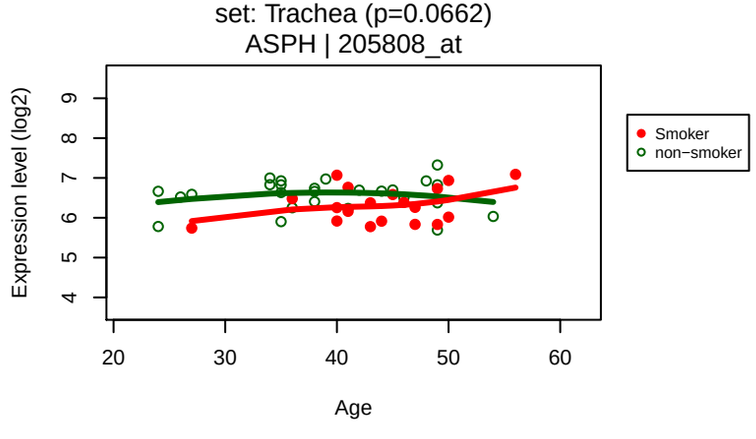

Gene expression ratio in Males
gene: ASPH | 205808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8448
p=0.5133
p=0.0898
p=0.0723
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASPH | 205808_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0679
p=0.0516
p=0.8218
p=0.8402
SmallAirway2
LargeAirway2
Trachea
Macrophage2

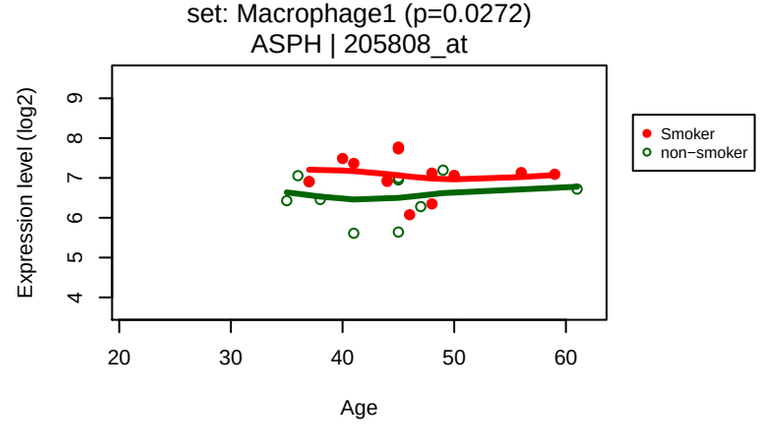

set: SmallAirway1 (p=0.002)
ASPH | 205808_at
Expression level (log2)
Age
Smoker
non−smoker

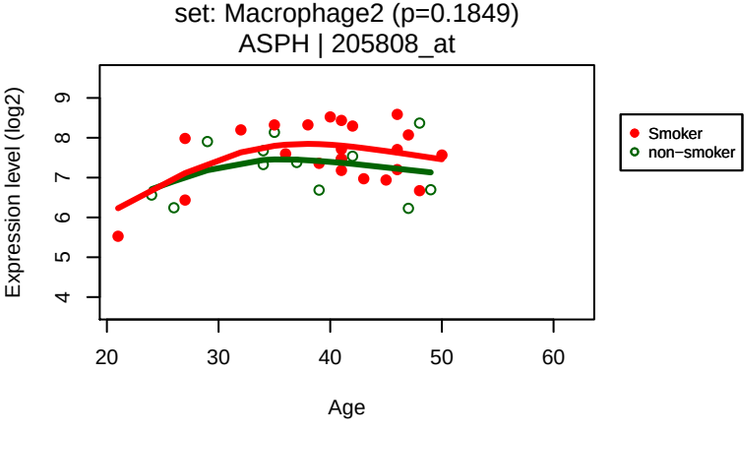

set: SmallAirway2 (p=0.3466)
ASPH | 205808_at
Expression level (log2)
Age
Smoker
non−smoker

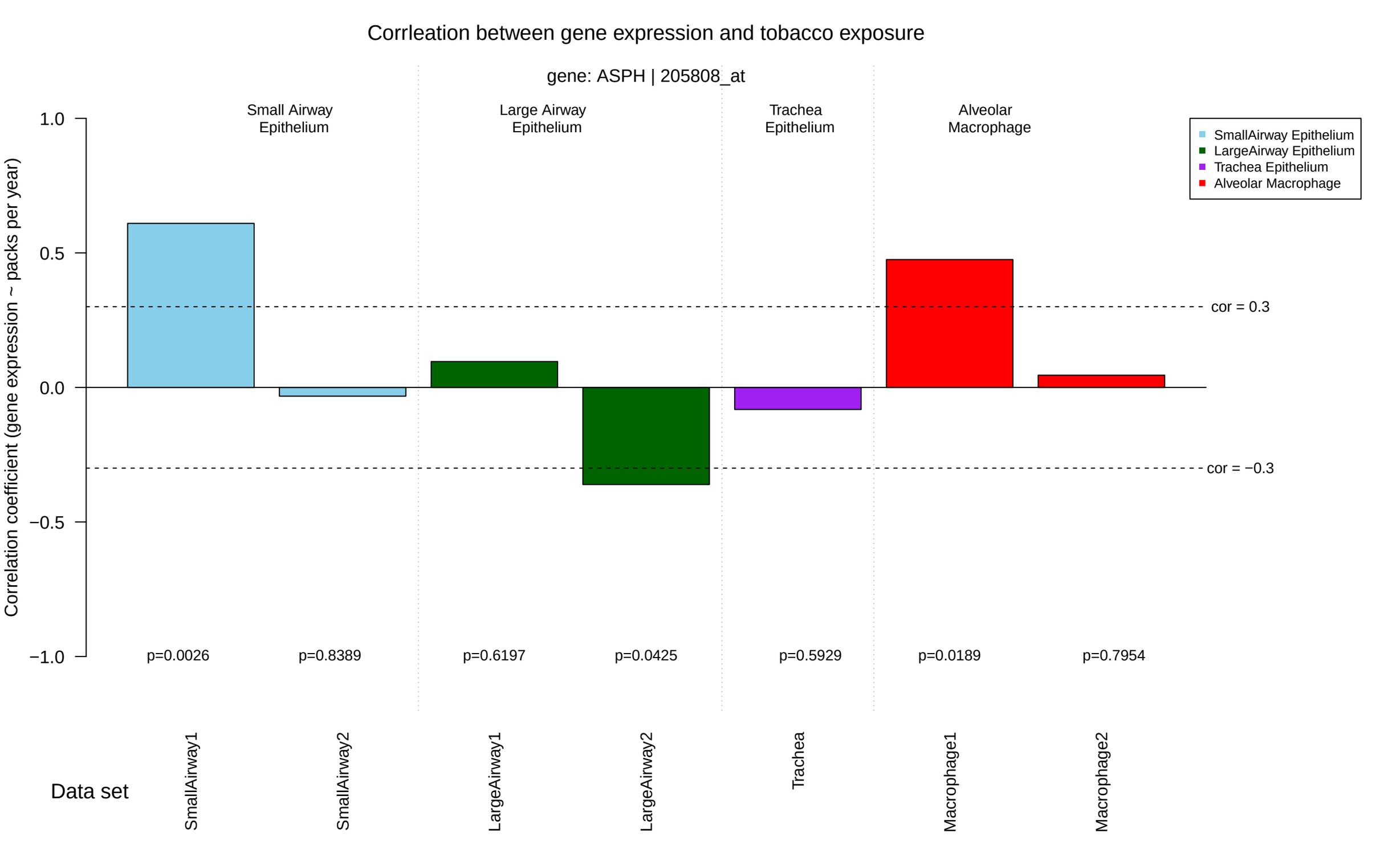

set: LargeAirway1 (p=0.3692)
ASPH | 205808_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.277)
ASPH | 205808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0662)
ASPH | 205808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0272)
ASPH | 205808_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1849)
ASPH | 205808_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASPH | 205808_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0026
p=0.8389
p=0.6197
p=0.0425
p=0.5929
p=0.0189
p=0.7954
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2